A valuation obstruction for 3×3 magic squares of squares, with a machine-checked proof

Florian Ernst Albin Morel

August 2026 — draft

Abstract

We study the open problem of whether a 3×3 magic square whose nine distinct entries are perfect squares exists. We reduce the problem to a purely additive condition on the set $D(e) = \{2xy : x^2 + y^2 = e^2\}$ attached to the center root e, and prove unconditionally that the center root of any such square must be divisible by at least two distinct primes $p \equiv 1 \pmod 4$: for centers $e = s\,p^a$ with a single such prime, the elements of $D(e)$ have pairwise distinct p-adic valuations, which is incompatible with the required additive structure. All results, including the reduction, are formally verified in Lean 4 with mathlib. We also report an exhaustive computation showing no magic square of squares has center entry below 10^{20}, and that Bremner's fully magic square with seven square entries is, up to scaling, the only one with center root below 10^9. Finally we describe a rigidity phenomenon that pins down the remaining two-prime case to explicit divisibility coincidences.

1 Introduction

Whether nine distinct perfect squares can form a fully magic 3×3 square is a well-known open problem, popularized by Martin Gardner's 1996 prize and studied extensively since; see Boyer's survey and problem collection [1]. The strongest results to date are computational (searches over magic sums and over arithmetic-progression parameters) together with structural constraints on hypothetical solutions due to Morgenstern, Roberts, Underwood and others [1].

Throughout, a *magic square of squares* is a 3×3 array of nine distinct perfect squares whose three rows, three columns and both diagonals share a common sum S. Summing the four lines through the center shows $S = 3E$ where $E = e^2$ is the center entry; we call e the *center root*.

Results

Theorem 1 (Reduction)**.** *Let $D(e) = \{2xy : x^2 + y^2 = e^2,\ 0 < x < y\}$. A magic square of squares with center e^2 exists if and only if there are u, v with u, v, $u + v$, $u - v \in D(e)$ (all nonzero, $u \neq v$); the square's entries are then $e^2,\ e^2 \pm u,\ e^2 \pm v,\ e^2 \pm (u + v),\ e^2 \pm (u - v)$.*

Theorem 2 (Main theorem)**.** *If the center root e is divisible by at most one prime $p \equiv 1$* (mod 4) *(to any power, and with arbitrary cofactor free of such primes), then no magic square* *of squares with center e^2 exists. Equivalently: the center root of any magic square of squares is divisible by at least two distinct primes $\equiv 1 \pmod 4$.*

Theorem 3 (Computations)**.** *There is no magic square of squares with center entry $e^2 < 10^{20}$. Moreover, every fully magic 3×3 square with exactly seven square entries and center root $e < 10^9$ is an integer scaling of Bremner's example with $e = 425$.*

All statements of Theorems 1 and 2, including Fermat's theorem that three squares in arithmetic progression have common difference divisible by 24, are formally verified in Lean 4 over mathlib; the library (thirteen files, no unproven assumptions) accompanies this paper.

2 The reduction

Every 3×3 magic square has the form $c \pm \{0, u, v, u+v, u-v\}$ arranged as

$$\begin{pmatrix} c+u & c-u-v & c+v \\ c-u+v & c & c+u-v \\ c-v & c+u+v & c-u \end{pmatrix},$$

in which all eight line sums equal $3c$ identically. Thus magic is free and the entire problem is the squareness of the nine cells.

Lemma 4. *For $c = e^2$, both $c-d$ and $c+d$ are squares if and only if $d \in D(e) \cup \{0\}$ (up to sign). Indeed if $x^2 + y^2 = e^2$ then $c - 2xy = (x-y)^2$ and $c + 2xy = (x+y)^2$; conversely if $c - d = A^2$ and $c + d = B^2$ then $A^2 + B^2 = 2e^2$, $A \equiv B \pmod 2$, and $x = \frac{B-A}{2}$, $y = \frac{B+A}{2}$ satisfy $x^2 + y^2 = e^2$, $2xy = d$.*

Theorem 1 follows by applying Lemma 4 to the four center lines. (Distinctness of the nine entries corresponds to $u, v, u \pm v \neq 0$ and $u \neq v$.)

3 Proof of the main theorem

Fix a prime $p \equiv 1 \pmod 4$, write $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$ with $\pi = A + Bi$, $A^2 + B^2 = p$, and let $e = s\,p^a$ with $a \geq 0$ and s free of primes $\equiv 1 \pmod 4$.

Lemma 5 (Rigid part)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2$ is a unit multiple of s.*

Proof. Induct on the least prime factor q of s. If $q \equiv 3 \pmod 4$ then q is inert; $q \mid N(z) = z\bar{z}$ gives $q \mid z$, and we divide. If $q = 2$, then $1 + i$ divides z twice (as $2 \mid N(z)$ twice), and $(1+i)^2 = 2i$ is 2 times a unit. □

Lemma 6 (Classification)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2 p^n$ equals $u\,s\,\pi^j \bar{\pi}^{n-j}$ for a unit u and some $0 \leq j \leq n$.*

Proof. Induct on n: for $n \geq 1$, $\pi \mid p \mid z\bar{z}$, so $\pi \mid z$ or $\bar{\pi} \mid z$; divide and recurse, finishing with Lemma 5. □

Lemma 7 (Bridge)**.** *For every $m \geq 1$, $p \nmid \operatorname{Im}(\pi^m)$.*

Proof. Pick $\omega \in \mathbb{Z}/p$ with $\omega^2 = -1$ (possible as $p \equiv 1 \bmod 4$) and consider the ring maps $\varphi_\pm : \mathbb{Z}[i] \to \mathbb{Z}/p$, $i \mapsto \pm\omega$. If $p \mid \operatorname{Im}(\pi^m)$ then, since $\operatorname{Re}(\pi^m)^2 + \operatorname{Im}(\pi^m)^2 = p^m$, also $p \mid \operatorname{Re}(\pi^m)$, so $\varphi_\pm(\pi)^m = 0$; as $\mathbb{Z}/p$ is a field, $\varphi_\pm(\pi) = 0$, i.e. $A \pm B\omega \equiv 0$. Adding and subtracting gives $p \mid A$, $p \mid B$, contradicting $A^2 + B^2 = p$. □

Lemma 8 (Representation structure)**.** *Let $x^2 + y^2 = e^2 = s^2 p^{2a}$ with $xy \neq 0$. Then there are $1 \le t \le a$ and $\varepsilon \in \{\pm 1\}$ with*

$$2xy \;=\; \varepsilon\, p^{\,2(a-t)} \left(s^2 \operatorname{Im} \pi^{4t}\right), \qquad p \nmid s^2 \operatorname{Im} \pi^{4t}.$$

In particular $\nu_p(2xy) = 2(a-t)$, and distinct elements of $D(e)$ have pairwise distinct p-adic valuations $\{0, 2, \ldots, 2a-2\}$.

Proof. Write $z = x + yi$; by Lemma 6, $z = us\pi^j \bar{\pi}^{2a-j}$. Then $2xy = \operatorname{Im}(z^2)$ and $z^2 = u^2 s^2 \pi^{2j} \bar{\pi}^{2(2a-j)}$ with $u^2 = \pm 1$. If $j = a$ the product is real and $xy = 0$. Otherwise, with $t = |j-a|$, $\pi^{2j}\bar{\pi}^{2(2a-j)} = p^{2(a-t)} \cdot \pi^{\pm 4t}$, and $\operatorname{Im}(\bar{\pi}^{4t}) = -\operatorname{Im}(\pi^{4t})$; absorb signs into ε. Lemma 7 and $p \nmid s$ give the non-divisibility. □

Proof of Theorem 2. Suppose $u, v, u+v, u-v \in D(e)$ as in Theorem 1, and give each the form of Lemma 8 with parameters (t_i, ε_i) and constants $c_t = s^2 \operatorname{Im} \pi^{4t}$. If $t_1 = t_2$ then $u = \pm v$, excluded. Otherwise $\nu_p(u) \neq \nu_p(v)$; writing $u \pm v = p^{\,\min} \cdot (\text{bracket})$ where the bracket is $\varepsilon_i c_{t_i} + (\text{multiple of } p)$, the bracket is prime to p. Comparing with the exact forms of $u + v$ and $u - v$ and using uniqueness of exact p-power extraction, $2(a - t_3) = 2(a - t_4) = \min$, so $t_3 = t_4$ and $u + v = \pm(u - v)$, forcing $u = 0$ or $v = 0$. Contradiction. □

Remark 9. The counting argument (four center lines require four representations) already forces two distinct primes *or* $p^4 \mid e$; the valuation argument closes the prime-power loophole entirely and is what we formalize.

4 Computations

Enumerating candidate centers directly through Theorem 1 is far cheaper than enumerating magic sums: factoring e in windows and generating $D(e)$ from the Gaussian factorization checks all $e < 10^{10}$ (center entries to 10^{20}) in minutes on a desktop; no configuration $u, v, u \pm v \in D(e)$ exists in that range. The same machinery classifies near-misses: configurations with two of the four differences in $D(e)$ and the other two "half in" (one square cell each) are exactly the fully magic squares with seven square entries, of which Bremner's $e = 425$ example was the only one known. Scanning all $e < 10^9$ finds 2,352,941 such configurations — every single one an integer scaling of $e = 425$, and none with eight square entries.

5 The minimal two-prime case

Theorem 10. *There is no magic square of squares whose center root is $s\,pq$, where $p \neq q$ are primes $\equiv 1 \pmod 4$, each to the first power, and s is free of such primes.*

Proof sketch. Here $D(e)$ has exactly four elements: the interior conjugate pair $m, m' = s^2|\operatorname{Im}(\pi^4\chi^{\pm 4})|$ and the axis elements $X = s^2 q^2 |\operatorname{Im} \pi^4|$, $Y = s^2 p^2 |\operatorname{Im} \chi^4|$. Since $u, v, u+v, u-v$

are four distinct elements of $D(e)$, they exhaust it. Write $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$; recall R_A, R_B are odd, $4 \mid I_A, I_B$, and by Lemma 7 $p \nmid I_A$, $q \nmid I_B$ (and $I_A, I_B \neq 0$). Enumerate the placements of $\{X, Y\}$ among $\{u, v, u+v, u-v\}$:

(1) $\{u, v\} = \{X, Y\}$. Then $\{u \pm v\} = \{m, m'\}$ and the conjugate-pair identities force $\{u, v\} = \{s^2|R_B I_A|,\ s^2|I_B R_A|\}$. Matching against X, Y and multiplying the two resulting equations gives $|R_A R_B| = p^2q^2$; but $|R_A| < p^2$ and $|R_B| < q^2$ strictly (as $I_A, I_B \neq 0$), a contradiction. The crossed matching forces $|R_A| = p^2$ outright, equally impossible.

(2) $\{u, v\} = \{m, m'\}$. Then $\{u \pm v\} = \{X, Y\}$ and $u+v,\ u-v \in \{2s^2|R_B I_A|,\ 2s^2|I_B R_A|\}$; matching yields either $2|R_B| = q^2$ or $4|R_A R_B| = p^2q^2$, both impossible by parity.

(3) mixed placements. One axis element lies in $\{u, v\}$, the other in $\{u \pm v\}$. Each of the 64 sign/placement sub-cases yields two linear equations in $W = R_A I_B$, $Z = I_A R_B$; solving and substituting into the norm relations $(W/I_B)^2 + I_A^2 = p^4$, $(Z/I_A)^2 + I_B^2 = q^4$ leaves a difference that factors, in every sub-case, into terms from the list

$$(I_A q^2 \pm I_B p^2), \quad (2I_A q^2 \pm I_B p^2), \quad (I_A q^2 \pm 2I_B p^2), \quad 3I_A q^2, \quad 3I_B p^2.$$

The all-positive combinations cannot vanish; the difference combinations vanish only if $I_A q^2 \in \{I_B p^2,\ 2I_B p^2\}$ or $I_B p^2 = 2I_A q^2$, each of which equates a quantity prime to p with one divisible by p^2 (Lemma 7). Hence no sub-case admits a solution. $\square$

Remark 11. Bremner's seven-square-entry example has center root $425 = 5^2 \cdot 17$, just outside the hypothesis (exponent 2): consistent with Theorem 10, the near-miss lives at the smallest center type the theorem does not reach.

6 Center roots $s\,p^2q$: Bremner's type

Theorem 12. *There is no magic square of squares whose center root is $s\,p^2q$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and s is free of such primes.*

Here $|D(e)| = 7$: writing $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$ and $P = p^2$, $Q = q^2$, the elements are (up to the global factor s^2)

$$P\,|\mathrm{Im}(\pi^4\chi^{\pm 4})|, \quad |\mathrm{Im}(\pi^8\chi^{\pm 4})|, \quad PQ\,|I_A|, \quad Q\,|\mathrm{Im}\,\pi^8|, \quad P^2|I_B|.$$

All seven are *linear* in (R_B, I_B) over $\mathbb{Q}(R_A, I_A, P, Q)$. The four differences $u, v, u+v, u-v$ occupy four distinct elements; for each of the 840 placements and each resolution of the absolute values (6,720 linear systems in all), solving the two placement equations for (R_B, I_B) and substituting into $R_B^2 + I_B^2 = Q^2$ leaves one polynomial relation in (R_A, I_A, P, Q), which must vanish. A computer-assisted but fully rigorous sweep (scripts accompany this paper) shows every relation is impossible:

- 339 of the 343 distinct relation classes die by combinations of: parity (R_A, R_B, P, Q odd, $8 \mid I_A, I_B$); the p-adic clash (ν_p of the two sides differ, using $p \nmid R_A I_A$ and Lemma 7); congruence obstructions modulo 16, 32 and small odd moduli on the curve $R_A^2 + I_A^2 = P^2$; the strict size pinch $|R_A|, |I_A| < P$; and elimination of I_A via the norm followed by factorization. The finitely many exceptional primes ($\ell \in \{5, 13, 17\}$, arising from small constants in the eliminants) are closed by exact evaluation at the actual π^4.

- 4 classes reduce to homogeneous quintics/sextics in (R_A, P); a rational root R_A/P in lowest terms would need $P = p^2$ to divide a leading coefficient in $\{\pm 8, \pm 16, \pm 32, \pm 40\}$ — impossible.
- 4 classes are tautological (the norm residual vanishes identically); there the solved values are $(R_B, I_B) = \pm(Q/P)(R_A, -I_A)$, and integrality forces $p^2 \mid q^2$.

The 256 distinct determinants of the linear systems are proven nonzero by the same mechanisms, so no singular case escapes. □

Remark 13. Bremner's fully magic square with seven square entries has center root $425 = 5^2 \cdot 17$, of exactly this type: Theorem 12 shows the missing two entries can never be repaired at *any* center of that shape. Since the seven D-elements are linear in (R_B, I_B) for every $b = 1$ exponent pattern $s\,p^a q$, the same pipeline applies to all $(a, 1)$ cases; the case count grows with a, and a uniform-in-a argument is the natural next objective.

Theorem 14. *There is no magic square of squares whose center root is $s\,p^3 q$ ($p \neq q$ primes $\equiv 1 \bmod 4$, s free of such primes).*

For $a = 3$ the difference set has ten elements, still linear in (R_B, I_B); the sweep comprises 64,512 leaf systems in 3,767 relation classes. All 1,271 nonsingular classes die by the Theorem 12 mechanisms; the 2,496 singular classes (rank-deficient placements, which first appear at $a = 3$ through the three constant axis elements) die through their consistency relations by the same kills; and all 1,080 determinant classes are proven nonzero, so no singular locus escapes. The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ share one pipeline, and every two-prime exponent pattern (a, b) is decidable in this sense — for $b \geq 2$ the leaf systems become zero-dimensional quadratic rather than linear, since $\chi^8, \chi^{12}, \ldots$ are polynomials in (R_B, I_B). The open problems are uniformity in the exponents and the ≥ 3-prime landscape, where the leaf systems acquire positive dimension.

7 Center roots $s\,p^3 q$

Theorem 15. *There is no magic square of squares whose center root is $s\,p^3 q$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and s is free of such primes.*

Here $|D(e)| = 10$: the seven Theorem 12 classes scaled by p^2, together with the three genuinely new elements $Q\,|\mathrm{Im}\,\pi^{12}|$ and $|\mathrm{Im}(\pi^{12}\chi^{\pm 4})|$. The formal proof (`TheoremGCore.lean`, again zero `sorry`s) mirrors the Theorem 12 architecture rung for rung, and the comparison is instructive for the eventual induction on the exponent:

- the all-low bucket of the 16-bucket router cancels p^2 and invokes the Theorem 12 router verbatim — the ladder telescopes;
- the mixed core, which at level 8 required a page of residue analysis, collapses at level 12 to a three-line valuation argument: every low class is divisible by p^2, so the paired equations force $p^4 \mid q^4$;
- the new $q^2\,\mathrm{Im}\,\pi^{12}$ class dies through the classification of $J = 4R_4^2 - p^4$ and $J' = 4R_4^2 - 3p^4$ against powers of q, with unit branches killed modulo 16 and by consecutive squares,

and one deep cell descending to $p^4 \mid 20t^2J^2$, which forces $p = 5$ and then $5 \mid tJ$, a contradiction;

- the ratio layer's hardest cell, $\mathrm{Im}(\pi^{12}\chi^4) = \pm 2\,\mathrm{Im}(\pi^{12}\bar{\chi}^4)$, factors through $\pi^{12} = (\pi^6)^2$ and runs the same Fermat descent as Theorem 12 with base pair (q, p^3) in place of (q, p^2).

The proof is thus not merely analogous to Theorem 12 but *reuses* it as a black box wherever a factor p^2 can be cancelled, and the three new classes are dispatched by arguments whose shape is independent of the exponent. This is precisely the inductive step one would want for a uniform theorem covering all center roots $s\,p^aq$.

8 Center roots $s\,p^aq$: the uniform theorem

Theorem 16. *There is no magic square of squares whose center root is $s\,p^aq$, for any $a \geq 1$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and s is free of such primes.*

This theorem subsumes Theorems **??**, 12 and 15 in one induction on a. The formal proof (`UniformAInt.lean`, zero `sorry`s) makes the telescoping of the concrete proofs exact. The class family $\mathrm{UClass}(a)$ has $1 + 3a$ members: rung a equals $p^2 \cdot \mathrm{UClass}(a-1)$ plus the three new classes $q^2\,\mathrm{Im}\,\pi^{4a}$ and $\mathrm{Im}(\pi^{4a}\chi^{\pm 4})$. The proof has four layers.

- *Generic cores and dispatchers.* The four cross-pair cores and the ten dispatchers of the concrete proofs hold at every rung a simultaneously. The generic arguments are simpler than the concrete ones: the unit branches die by size and by consecutive squares, and the residue $\pm q^2$ plus one congruence kills each partner class.

- *The assignment step.* The sixteen-bucket router at rung a takes the rung $a-1$ router as a hypothesis. Its all-low bucket cancels p^2 and recurses; the other buckets route into the generic dispatchers.

- *The ratio step.* The ratio-2 router telescopes the same way. Its hardest cell, $\mathrm{Im}(\pi^{4a}\chi^4) = \pm 2\,\mathrm{Im}(\pi^{4a}\bar{\chi}^4)$, factors through $\pi^{4a} = (\pi^{2a})^2$ and runs the Fermat descent with base pair (q, p^a), generic in the odd base.

- *The base.* At rung 1 the four classes are concrete; two low classes cannot fill four distinct slots, and the generic dispatchers close the remaining buckets.

The induction then assembles both routers for every $a \geq 1$, and the four-differences argument and the center reduction give the theorem.

9 Center roots $s\,p^2q^2$

Theorem 17 (Theorem H, paper-level)**.** *There is no magic square of squares whose center root is $s\,p^2q^2$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and s is free of such primes.*

This is the first case with both exponents ≥ 2. Here $|D(e)| = 12$, on the grading grid $p^{4-2\alpha}q^{4-2\beta}\,|\mathrm{Im}(\pi^{4\alpha}\chi^{\pm 4\beta})|$ with $\alpha, \beta \in \{0, 1, 2\}$. The leaf enumeration has 95,040 assignments. The proof has four layers; each is uniform in p and q, and each is validated numerically over the full prime grid.

1. *Grading.* A lone minimal p- or q-layer kills 93.5% of the leaves (88,832), leaving 6,208.

2. *Unit certificates.* For a surviving leaf, both relations must vanish, so p divides both minimal p-layers. Reduce both mod π in the orbit basis ($R \equiv \bar{\pi}^4/2$, $I \equiv i\bar{\pi}^4/2$, $V = q^4/U$ with $U = \chi^4$) and take the resultant in U. For 5,088 leaves the resultant is a pure unit monomial, such as $16\,\bar{\pi}^{32}q^{32}$: simultaneous vanishing would force $\pi \mid 16$. The q-side pass kills 864 more. Remaining: 256 leaves, degenerate on both sides.

3. *Strict pinch.* Of these, 192 factor over $\mathbb{Z}$ with a factor $R \mp p^2$, $X \mp q^2$, $R^2 - I^2 \mp p^4$ or $X^2 - Y^2 \mp q^4$: the real part of a power of π or χ never reaches its norm while the imaginary part is nonzero. Remaining: 64 leaves in four conjugate-pair families.

4. *The master quartic kill.* Every remaining relation states, with $u = \pi^2$, $v = \bar{\pi}^2$, $x = \chi^2$, $y = \bar{\chi}^2$, that $\chi^8 B$ (or symmetrically $\pi^8 C$) is purely real or purely imaginary, where B is one of the quartics $u^4 + v^4 \pm 2u^3v$ and their mirrors (verified for all 128 relations by exact multi-point evaluation). Real or imaginary balance forces equal conjugate valuations, so $\bar{\chi}^8 \mid B$ and $B = d\,\bar{\chi}^8$ with $d \in \mathbb{Z} \cup i\mathbb{Z}$, $d \neq 0$. Taking norms, with $(r,s) = (\operatorname{Re}\pi^2, \operatorname{Im}\pi^2)$, $r^2 + s^2 = p^2$:

$$N(B) \;=\; 16s^2\big[(3r^2 - s^2)^2 s^2 + r^2p^4\big] \;=\; |d|^2 q^8,$$

so $w^2 = \big(s(3r^2 - s^2)\big)^2 + (rp^2)^2$ for some $w \in \mathbb{Z}$ — a Pythagorean condition with legs $\operatorname{Im}(\pi^6)$ and $\operatorname{Re}(\pi^2)\,p^2$. The triple is primitive (r odd, s even, all pairwise gcds 1), so $rp^2 = (m-n)(m+n)$ with $m-n$, $m+n$ coprime. Since $m - n \leq |r| < p < p^2$, the factor p^2 sits whole in $m+n$; writing $|r| = r_1r_2$ accordingly, $2mn = r_2^2p^4 - r_1^2 \geq p^4 - p^2$, while $2mn = 2|s|\,|4r^2 - p^2| < 6p^3$. Hence $p \leq 6$, so $p = 5$; there $(r,s) = (\pm 3, \pm 4)$ gives $M = 7561$, not a square. Finally $B \neq 0$: from $B = 0$, taking absolute values in $u^3(u \mp 2v) = -v^4$ gives $|u \mp 2v| = p$, and squaring, $5p^2 \mp 4\operatorname{Re}\pi^4 = p^2$, so $\operatorname{Re}\pi^4 = \pm p^2$ — the strict pinch again, impossible while $\operatorname{Im}\pi^4 \neq 0$. No leaf survives.

The four-differences assembly and the center reduction then give the theorem, exactly as in Theorems 12–16. Two remarks. The unit certificates and the master kill never see the band loci that stopped the earlier leaf battery: the coupled disc-form coincidences satisfy the form-level conditions but never the exact divisibility coupling. And the master kill uses only π-side data, so it is already generic in the χ-exponent — the shape the two-exponent induction needs.

10 Toward all exponents: the balance-form reduction

The programme's next target is the two-exponent theorem: no center root $s\,p^aq^b$ for any $a, b \geq 1$. Three structural facts, each verified computationally at scale, fix its architecture.

First, *saturation in a*. Classify each surviving leaf by the pattern of its minimal p-layers (the classes and signs at the tying level, shifted to a common base). Over the grids up to $(5,5)$ — 93,625,920 leaves — the pattern universe never grows with a beyond $a = 2$, at any b; it grows only with b, linearly. So the certificates live on the b-axis, and every cell argument is automatically generic in a.

Second, *the balance-form dichotomy.* At the first new rung $(2, 3)$, every leaf dies at one of four gates: the unit-certificate resultants; the nonzero-factor screen (variable monomials, $u^j \pm v^j$-type coordinate forms, and strict pinches); a lone-minimal-valuation monomial with coefficient ± 1 (the prime would divide 1); or the remaining factors are *balance forms* — polynomials F with $F = \pm \bar{F}$ under the simultaneous swap $u \leftrightarrow v$, $x \leftrightarrow y$, so that their value is forced real or purely imaginary. All 1,120 unresolved factors at $(2, 3)$ are balance forms or unit-graded; none is a genuine core. No balance factor vanishes at any point with $p \neq q < 300$ (3.6 million exact evaluations).

Third, *the balance lemma.* Write a balance form's positive half as $G = \sum_t c_t \, \pi^{2a_t} \bar{\pi}^{2b_t} \chi^{2e_t} \bar{\chi}^{2f_t}$ with $0 < |c_t| < q$. If $\operatorname{Im} G = 0$ (or $\operatorname{Re} G = 0$), then $v_\chi(G) = v_{\bar{\chi}}(G)$. Every balance factor in the catalog has *clean* minimal layers: the terms at the minimal χ-valuation form a single monomial $\chi^{2e^*} \bar{\chi}^{2f} P(\pi^2, \bar{\pi}^2)$ (verified for all 1,074 shapes, on both sides). When the profile is shifted ($e^* < f^*$), the valuation balance forces $\chi^{2\Delta} \mid P$-value with $\Delta = f^* - e^*$, hence

$$q^{2\Delta} \;\Big|\; N(P),$$

an explicit integer form in (p, s), $s = \operatorname{Im} \pi^2$. The computed $N(P)$ factorizations are benign throughout the catalog: products of p-powers and small forms such as $p \pm s$, $p \pm 2s$, $(p^2 - 2s^2)^2$, $(3p^2 - 4s^2)^2$ (the identities $p^2 \pm \operatorname{Re} \pi^4 = 2r^2, 2s^2$ collapse several families). Each divisibility gives a band $q^\Delta \leq Cp^k$ plus an exact small-factor condition; on the residual thin loci the full balance pins an exact norm equation, and the Theorem 17 splitting argument is the prototype finish. The finish is in fact sharper than the norm divisibility, through the following exactness lemma.

Lemma 18 (Exactness)**.** *Let $M = \chi^{2e} \bar{\chi}^{2f} P$ with $P = P(\pi^2, \bar{\pi}^2) \neq 0$ and $\Delta = f - e > 0$. If M is real, then $\chi^{2\Delta} \mid P$-value, and $P = \tilde{w} \, \chi^{2\Delta}$ with $\tilde{w} \in \mathbb{Z}$; if M is purely imaginary, the same holds with $\tilde{w} \in i\mathbb{Z}$.*

Proof. $M = \bar{M}$ reads $\chi^{2e} \bar{\chi}^{2f} P = \chi^{2f} \bar{\chi}^{2e} \bar{P}$; dividing by $\chi^{2e} \bar{\chi}^{2e}$ gives $\bar{\chi}^{2\Delta} P = \chi^{2\Delta} \bar{P}$. Since $\chi \nmid \bar{\chi}$, unique factorization gives $\chi^{2\Delta} \mid P$; writing $P = \chi^{2\Delta} \kappa$ and substituting yields $\bar{\chi}^{2\Delta} \kappa = \bar{P} = \overline{\chi^{2\Delta} \kappa} = \bar{\chi}^{2\Delta} \bar{\kappa}$, so $\kappa = \bar{\kappa} \in \mathbb{Z}$. The imaginary case is identical with $M = -\bar{M}$. ☐

For a balance factor with several asymmetric layers the same cascade applies level by level (the catalog factors have clean minimal layers, so each step is an instance of the lemma), and the layered factors follow the q^{2d}-division chain described below. The exact condition, together with its conjugate, *linearizes*: writing $\Sigma = P + \bar{P}$ and $\Omega = (P - \bar{P})/2i$ (explicit integer forms in the π-coordinates),

$$2\tilde{w} \, \operatorname{Re} \chi^{2\Delta} = \Sigma, \qquad \tilde{w} \, \operatorname{Im} \chi^{2\Delta} = \Omega, \qquad 4\tilde{w}^2 q^{2\Delta} = \Sigma^2 + 4\Omega^2.$$

The master forms $\Sigma^2 + 4\Omega^2$ are computed for all 58 families: small explicit quartics such as $4(8S^2 + p^2)$ and $16(8S^4 - 5S^2 p^2 + p^4)$. Any prime of $\tilde{w}$ divides both Σ and Ω, and the coprimality of the π-coordinates confines $\tilde{w}$'s support to a fixed small integer per family (in the prototype family $P = v(v - 2u)$: $\tilde{w} = \pm 3^t$; the families with coefficients $(2, \pm 1)$ reproduce the S-core quadratic $2(4R^2 \mp 3p^2 R + p^4)$ of the uniform theorem, and the families with $\alpha = \pm \beta$ die outright since $\tilde{w} \, \operatorname{Im} \chi^{2\Delta} = 0$ forces $\operatorname{Re} \pi^{2m} = 0$). Thus each balance condition pins the *coordinates* of $\chi^{2\Delta}$ to explicit π-side forms divided by one of finitely many $\tilde{w}$. A surviving leaf carries two balance conditions, which pin the same coordinates to two different forms;

eliminating χ leaves a pure π-side Diophantine system on the circle $R^2 + S^2 = p^{2m}$, and the band and Pell arguments of the earlier sections close it. The complete sector map of rung $(2,3)$ is now mechanically certified. A condition with exactly one of Σ, Ω identically zero is impossible outright (it forces Re or Im of $\chi^{2\Delta}$ to vanish). Monomial families are impossible by pure valuation (χ-content in a π-power value). For two live conditions with equal shifts, the compatibility cross $\Sigma_1\Omega_2 - \Sigma_2\Omega_1$ is $\tilde{w}$-free, and every one of its factors is parity-excluded or has only irrational roots. For unequal shifts, the two pinnings force q-power divisibility of $A_{12} = \Sigma_1\Sigma_2 + 4\Omega_1\Omega_2$ and of the cross; the resultant contents are powers of 2, and the polynomial gcds are powers of $r^2 + s^2 = p^2$ — both impossible for $q \geq 5$, $q \neq p$. With these filters, the entire non-layered sector of $(2,3)$ is dead; notably, the disc-8 Pell coincidences that stopped the earlier $(2,2)$ program appear only on branches that are already dead.

The remaining sector is *layered*: factors of the sandwich form $\bar{G} + D + G$ whose diagonal (real) layer sits below the asymmetric minimum, in fourteen profiles. Dividing the three-term equation by q^{2d} yields the chain: $\chi^{2d} \mid \bar{P}$-value (a band through the computed norm formulas; monomial P dies here by valuation), then an inhomogeneous master condition $\mathrm{Im}(\bar{\chi}^{2(\Delta-d)}W) = -cT$ with $W = \bar{P}/\chi^{2d}$ and T an explicit π-side integer from the diagonal layer — ninety-five distinct systems over the same small P-catalog, with the partner relation's cross still available on top. The assembly is a q^2-telescoping induction in b, in the exact image of the uniform theorem's induction in a. At rung b the class family splits as $q^2 \cdot (\text{rung } b-1)$ plus the top classes $(j, b, \pm)$, $j \leq 2$. Leaves with all four classes low cancel q^2 from both E-relations and recurse; leaves with top classes fall into the certified sectors, whose kills involve the shift Δ only through the divisor $q^{2\Delta}$ and its band — and both only tighten as Δ grows, so every cell argument is monotone in b. The base of the induction is Theorem 17; the row and column of exponent one are Theorem 16 and its swap; and the saturation census transports each cell from $a = 2$ to all a. The certifier (the pipeline reimplemented in Rust: library division in place of general factoring, the affine model, Bareiss resultants over the χ^2-circle) closes the grids $(2,3)$, $(2,4)$, $(2,5)$, $(3,3)$ and $(3,4)$ completely, in seconds to minutes each. Exactly one cell resisted the mechanical layers, the *aligned* family at $a = 3$: there $\Sigma = \pm 2p^4$ and $\Omega = \pm 2\,\mathrm{Im}\,\pi^8$, the smooth $\tilde{w}$ divides p^4, and the pinning is the exact identity $\chi^{2\delta} = \pm p^4 \pm (\pi^8 - \bar{\pi}^8)$, i.e. $q^{2\delta} = R_8^2 + 5S_8^2$. The finish is elementary: with $W = q^\delta$, the coprime splitting of $(W - R_8)(W + R_8) = 5S_8^2$ (the gcd divides 5, and 5 is excluded by $\gcd(R_8, S_8) = 1$) gives $S_8 = 2\alpha\beta$ and $p^8 = \beta^4 - 6\alpha^2\beta^2 + 25\alpha^4 = N\big((\beta^2 - 3\alpha^2) + 4i\alpha^2\big)$; the Gaussian integer $z = (\beta^2 - 3\alpha^2) + 4i\alpha^2$ is odd and coprime to its conjugate, so $z = u\,\pi^8$; comparing imaginary parts with $S_8 = 2\alpha\beta$ forces $\beta = \pm 2\alpha$ (for $u = \pm 1$), hence $\alpha = 1$ and $p^8 = 17$, absurd, while $u = \pm i$ makes R_8 even, absurd. The cases $p = 5$ and $q = 5$ die by direct divisibility. With this cell closed, every swept grid is entirely dead.

The passage from the swept grids to all (a, b) is now a finite, δ-free bookkeeping. Write each live condition's pinning as $\chi^{2\delta} = z/\tilde{w}$ with $z = \Sigma/2 + i\,\Omega$ an explicit π-side Gaussian integer and $\tilde{w}$ in the finite smooth set; the shift δ enters nowhere else. Then: (i) for twenty-eight of the fifty-eight families the master norm $N(z) = \Sigma^2/4 + \Omega^2$ carries a mandatory factor p, and $\tilde{w}^2 q^{2\delta} = N(z)$ forces $p \mid \tilde{w}$ — dead in one line, for every δ; (ii) two conditions at distinct shifts $\delta_1 \neq \delta_2$ have master forms whose polynomial gcd (computed over all 1,653 family pairs) is a smooth constant times powers of p, s, $p \pm s$, except for a handful of pairs sharing the forms $p^2 + 8s^2$ or $9p^2 - 8s^2$; in the coprime cases the q-parts of the two masters cannot both be nontrivial, and in the shared-form cases dividing the two master equations gives $q^{2(\delta_1 - \delta_2)}$

equal to a smooth times a p-power, forcing $\delta_1 = \delta_2$ — either way the unequal-shift case dies or collapses; (iii) two conditions at the same shift are killed by the cross $\Sigma_1\Omega_2 - \Sigma_2\Omega_1$, a δ-free polynomial whose nonvanishing is a finite verified list, unless the pair is aligned, in which case $\tilde{w}_2 z_1 = \pm\tilde{w}_1 z_2$ is itself an explicit π-side identity, nonvanishing except in the proportional case — and the full catalog scan shows the proportional case is exactly the $\sqrt{-5}$ family, killed above. Threaded through the q^2-telescoping induction with the swept grids as bases, this closes every (a, b); the remaining rigor debt is the exact-root oracle pass on the certifier's final forms, running now with all completed grids reporting every form verified.

11 The two-prime frontier

Toward uniformity in the exponent

Theorem 16 now closes the family $s\,p^a q$ for all a at once. The analytic program below preceded that proof; it retains independent interest, because its tools aim past the $b = 1$ column toward the $(a, b \geq 2)$ grids. The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ suggested the uniform theorem. Abstracting the levels of $D(e)$ into free circle variables linked by a chain map (the enriched relaxation), 95.8% of the 9,216 uniform two-level leaf systems die by mechanisms independent of a — positivity pinches, parities, p-adic patterns and congruence obstructions. The residue is exactly 24 relations, independent of a, involving two π-powers π^{4j}, $\pi^{4(j+\Delta)}$. A complete weight-regime analysis shows that their phase-aligned leading forms cancel identically: the obstruction survives every first-order p-adic tool. The residue was then resolved further by the *exact orbit substitution*: under the two p-adic embeddings, $R_j - \omega I_j = \pi_-^{4j}$ exactly, so each relation becomes a finite Laurent polynomial with definite monomial valuations. Single-level templates die completely (unit leading coefficients; the tautological placements force $\operatorname{Im}\chi^4 = 0$; the $p = 5$ shapes die by exact 5-adic evaluation over a full period). Two-level templates die except on six explicit loci $\pi_+^{4\Delta} \equiv c \pmod{p}$, $c \in \{\pm 1, \pm 2, \pm\frac{1}{4}\}$; since $\pi_+^{4\Delta} = c$ exactly is impossible (norms), each locus kills all but finitely many j per (p, Δ), and effective bounds on $v_p(\pi_+^{4\Delta} - c)$ — p-adic linear forms in logarithms — yield $j \ll \log^2 \Delta$: a complete effective-finiteness program for the uniform theorem over all $s\,p^a q$, with only the Baker-bound bookkeeping, the resulting finite verification, and the three- and four-level chain templates outstanding.

Three and four levels: the Q-collapse and the leading pair

The three- and four-level chain templates are now closed to the same standard. A merged sweep (36,288 three-level and 31,104 four-level leaves) kills 95.7%, resp. 91.4%, by the zero-solution, imbalance, residual-unit and singular mechanisms. The residue collapses to 8, resp. 12, *deep shapes* — same-orientation interior elements plus one axis element — and for each of the twenty shapes a symbolic computation gives the decisive structural fact: writing the solved value as $\chi^4 = Q\,N/D$, the numerator has Q-degree exactly one and the denominator is Q-free. Setting $N = Q\,A$, the modulus condition $|\chi^4| = Q^2$ eliminates q *entirely*: a deep-shape leaf survives if and only if

$$c \;=\; |A|^2 - |D|^2 \;=\; 0,$$

a single integer identity in the π-orbit algebra. Expanding c over its monomial valuation groups $\gamma\,\pi^{P\cdot j}\bar{\pi}^{B\cdot j}$ (the substitution is exact, with no unit ambiguity), realness pairs the groups conjugately, and for *every* shape and *every* strong-separation ordering of the orbit indices the minimal-valuation set is exactly one conjugate pair with real coefficient $\gamma \in \{\pm1, \pm2\}$ — a unit for every odd p. The pair sums to $2\gamma\, p^{a_0}\,\mathrm{Re}(\bar{\pi}^{4M})$ and $v_p\big(\mathrm{Re}(\bar{\pi}^{4M})\big) = 0$, so $v_p(c)$ is finite: *every deep-shape leaf whose orbit indices avoid an explicit finite union of valuation-tie hyperplanes is impossible, uniformly in $p \geq 5$.*

On the tie set, layering c by π-exponent reduces everything to integer *phase sums* $S_e(j) = \sum_i \gamma_i (2A)^{B_i\cdot j}$ (here $\pi = A + Bi$): survival requires $p \mid S_\pi(j)$ and $p \mid S_{\bar{\pi}}(j)$ simultaneously — a double multiplicative-order condition on $2A \bmod p$. A census over $p < 200$, orbit indices ≤ 6, finds 1,806 such double-vanishing points among 362,880 (0.5%); exact evaluation shows $c \neq 0$ at every one, with even escape depths $v_p(c) - e_{\min} \in \{2, \ldots, 20\}$ governed by the multiplicative order of $2A$ — the special-value loci in exact arithmetic form. Independently, a direct sweep of the criterion over all twenty shapes ($p < 200$, indices ≤ 4, $q < 500$; 3,236,352 exact checks) finds no survivor. The uniform program for $s\,p^a q$ thus stands closed at one and two levels unconditionally, and at three and four levels outside the thin double-vanishing set, where the small-order degeneration awaits its own (expectedly self-similar) leading-pair analysis.

The tie-structure dichotomy

The hyperplane residue itself has now collapsed. Certifying the valuation-group geometry by exact rational linear programming (a phase-one simplex over $\mathbb{Q}$, no floating point), one finds for *every* one of the twenty deep shapes: every π-weight form carries a single group; no three forms can simultaneously achieve the minimum; and exactly *one* pair of forms is jointly-minimal-feasible — each shape owns a single tie hyperplane. On it, conjugation symmetry makes the minimal $\bar{\pi}$-layer sum to $(\gamma_1 + \gamma_2)\,t^e$ with $t = 2A \bmod p$. The certified coefficient pairs split the shapes in two:

- **Eight shapes have** $(\gamma_1, \gamma_2) = (1, 2)$: the layer is $3t^e \not\equiv 0 \pmod p$ for every $p \geq 5$, so $v_{\bar{\pi}}(c)$ is finite always and $c \neq 0$ — *these eight shapes are impossible unconditionally: all $p \geq 5$, all orbit indices, all q.*

- **Twelve shapes have** $(\gamma_1, \gamma_2) = (-1, 1)$: the layer vanishes identically, and the surviving locus is exactly the tie hyperplane, where the leading contribution is $\pi^e \bar{\pi}^{\,b}(\bar{\pi}^{\,\Delta b} - 1)$ and lifting-the-exponent gives the exact escape depth $v_\pi(\bar{\pi}^{\,d} - 1) + v_p(\Delta b/d)$ when $d = \mathrm{ord}_p(2A)$ divides Δb (and death otherwise).

A census over $p < 200$, indices ≤ 6 confirms the classification exactly: all 1,806 double-vanishing points lie in the twelve $(-1, 1)$-shapes, every one is the trivial cancellation (no mod-p accidents anywhere), and $c \neq 0$ at each.

In fact the $(-1, 1)$ cancellation is *exact*: the tie $P\cdot j = B\cdot j$ forces both weight coordinates of the pair to coincide, so the two terms annihilate as Gaussian integers and c equals the sum of the remaining groups on the locus. Running the leading analysis *recursively* — delete each exactly-cancelling pair, add its equality, and re-analyze on the sub-locus, terminating by dimension exhaustion (observed depth ≤ 2) — closes every branch of every shape except finitely many explicit *order loci* (about nineteen in total across the twelve shapes): ties of opposite unit coefficients in a single coordinate with distinct exponents $b_1 \neq b_2$. There the layer sum is $\gamma(t^{b_1} - t^{b_2})$ with $t = 2A \bmod p$, zero precisely when $\mathrm{ord}_p(2A)$ divides $b_1 - b_2$, in which case the valuation deepens by exactly $v_\pi(\bar\pi^{\,b_1-b_2} - 1)$ — a Fermat-quotient (Wieferich-type) quantity. Survival would require this deepening to outrun every remaining layer; the exact censuses refute it at every point in range. This is the complete structural characterization of the uniform $s\,p^a q$ frontier at three and four levels: finitely many Wieferich-type loci per shape, and everything else proven impossible.

Both exponents two: the $(2, 2)$ program

Theorem 17 now closes this case. The record below is the earlier program; its residual band loci were survivors of that method's toolset, and Theorem 17's certificates bypass them. For $e = s\,p^2q^2$ the set $D(e)$ has exactly twelve elements with a clean two-dimensional grading,

$$D(e) \;=\; \bigl\{\, p^{4-2\alpha} q^{4-2\beta} s^2 \,\operatorname{Im}\!\bigl(\pi^{4\alpha}\chi^{\pm 4\beta}\bigr) : (\alpha,\beta) \in \{0,1,2\}^2 \setminus \{(0,0)\}\,\bigr\}.$$

Of the 95,040 placement-and-sign leaves, 93.5% die by the *two-prime grading lemma*: a relation whose minimal p-layer (or q-layer) holds a single term is impossible, since that term is a p-unit (modulo π, $\operatorname{Im}(\pi^{4\alpha}\chi^{4\beta})$ reduces to a unit multiple of $\bar\pi^{4\alpha}\bar\chi^{4\beta}$) standing against terms divisible by two more powers of p. The 6,208 residual leaves collapse to 216 distinct three-term relations, and an exact sweep over every ordered pair of primes $p \neq q \equiv 1 \pmod 4$ below 1000 (39,234,560 checks) finds none satisfiable: *no magic square of squares has center root $s\,p^2q^2$ with $p, q < 1000$, for any s.* Beyond the sweep, per-factor certificates close a majority of the residual structure *unconditionally*: each relation factors over $\mathbb{Q}$, and a factor is proven nonzero by parity (always-odd integers), by a *unit side* (its reduction modulo π or χ factors into unit atoms times a nonzero constant c, so no prime beyond $|c|$ ever divides it — and $p \nmid F$ alone gives $F \neq 0$), or by a *double pinch* (each side reduces to units times a single small core $2\operatorname{Re}(\chi^2)$ or $2\operatorname{Im}(\chi^2)$, strictly bounded by twice the opposite prime and nonzero for an odd prime; the two prime orderings then close both cases with no wedge). Working in the full coordinates $\chi = C + Di$ sharpens this again: the cyclotomic cores factor further ($\operatorname{Re}\chi^4 = (C^2 - 2CD - D^2)(C^2 + 2CD - D^2)$), a prime dividing a product divides a single core, and cores of half-degree $\frac{1}{2}$ (such as $C \pm D \leq \sqrt{2q}$) bound one prime by roughly the square root of the other; chaining the two sides bounds *both* primes. All 110 relations of this type have explicit bounds at most 144 — inside the exhaustive sweep — so with the earlier certificates 134 of the 216 relations are dead and 98.35% of *all* 95,040 placement leaves are closed unconditionally. The 1,572 surviving leaves live in 82 relations: 18 reduce to explicit Pell-band equations (core $= \pm kp$ inside a constant prime-ratio band), 64 carry genuine exponent-≥ 2 wedges, and 24 have a vanishing first-order side. For three primes the same analysis profiles all 600 residual relations: 116 (those with all three side exponents ≤ 1) are impossible outside constant prime-ratio bands, and the remainder confine any survivor to explicit power-law prime families ($r \sim p^2$-type rays of the associated log-inequalities) — a complete geometric characterization of the three-prime frontier.

All thirty-two pure double-pinch relations are now *machine-checked dead* in Lean 4 with no `sorry`. The mechanism is packaged as a reusable toolkit: `pi_core_disj` proves that $\pi \mid (\varepsilon_2 q^2 + \varepsilon_3 \bar\chi^4)$ forces $p \mid 2(C^2 - D^2)$ or $p \mid 4CD$; `sq_core_disj` is the symmetric χ-side statement; and `double_pinch_finish` shows any combination of the two core disjunctions is absurd by the strict two-ordering pinch (trichotomy on p versus q). The thirty-two relations organize into four slot-triple families, each formalized by a parametric theorem over all sign choices $\varepsilon_i \in \{\pm 1\}$, all odd prime pairs, and all Gaussian representations: vanishing makes the Gaussian value self-conjugate, each prime's side subtracts its manifestly divisible terms from the conjugated relation (one conjugation class per family requires regrouping the six-term identity $G - \bar{G} = 0$ before the extraction), unit factors are stripped by primality, and the exposed integer cores feed the pinch. The remaining twenty-four unconditionally-killed relations all share a twin structure — a lone slot plus a repeated slot in both conjugations — under which $\mathrm{Im}(zw \pm z\bar{w}) = 2\,(\mathrm{Im}\,/\,\mathrm{Re}\ z)(\mathrm{Re}\,/\,\mathrm{Im}\ w)$ collapses the relation value over $\mathbb{Z}$ into a nonzero unit factor ($\mathrm{Im}\,\pi^4 = 4AB(A^2 - B^2)$-type) times an odd cofactor $(2m - p^a q^b)$, with no Gaussian divisibility required; twelve parametric twin-family theorems dispose of them. **All fifty-six relations killed unconditionally by the symbolic census at exponent $(2,2)$ are therefore machine-checked**, over all odd prime pairs, all Gaussian representations, and all sign variants — the unconditional layer of Theorem H$'$ for $s\,p^2q^2$ centers is fully formal.

Moreover, both certificate classes admit *exponent-uniform* formalizations. The twin collapse never uses the structure of the π-side factor, so four theorems with an opaque Gaussian parameter z and an opaque odd coefficient c kill every lone-plus-twin shape at every exponent (a,b) at once. And the four double-pinch families are one corner-triple shape $\{(\alpha-1,\beta),(\alpha,\beta-1),(\alpha,\beta)\}$ at different anchors — the common $p^{2u}q^{2w}$ coefficient factors out of the imaginary part — so two theorems with free anchor exponents (`uniform_corner_aligned`, `uniform_corner_mixed`) subsume all four families and their counterparts at every (a,b). The mixed conjugation class provably occurs only at π-anchor $\alpha = 1$: at larger anchors the χ-side extraction lands on the large core $e_3\pi^{4\alpha-2} - e_1\bar{\pi}^{4\alpha-2}$ and the size pinch genuinely fails, in exact agreement with the census.

Running the certificate engine at higher exponents confirms the uniform layer is not merely a lower bound: at $(3,2)$ the census kills 86 of 468 residual relations unconditionally, at $(3,3)$ it kills 132 of 1008 — and *every single kill at both exponents is an instance of the six uniform theorems*. The corner anchors with $\alpha \geq 2$ are all aligned, the $\alpha = 1$ anchors split aligned/mixed exactly as proved, and the new twin shapes (lone slots $(3,0)$ and $(0,3)$) have twin power a multiple of the lone power, which the final generalization (`uniform_twin_pi`/`uniform_twin_chi`, using $\mathrm{Im}((\pi^4)^k) \neq 0$ for all $k \geq 1$) covers. Pushing the census to $(4,2)$, $(4,3)$, $(5,2)$ and $(4,4)$ (124/816, 190/1752, 162/1260 and 272/3040 unconditional kills respectively) exposed two further twin phenomena, both now proved: odd lone powers admit *arbitrary* twin powers (the cofactor $u_k = \mathrm{Im}(z^k)/\,\mathrm{Im}(z)$ satisfies $u_k \equiv kR^{k-1} \bmod 2$), and lone powers divisible by four admit odd twin powers (a 2-adic gap: $v_2(u_k) = v_2(k)$ exactly, so the collapsed cofactor $2e\,u_T\,\mathrm{Re}(w) - c\,u_L$ is $2 \cdot \mathrm{odd}$). The resulting *twin master rule* — the collapse kills if and only if $v_2(L) \neq 1 + v_2(T)$ — organizes every observed twin pair, and the pairs that would violate it never occur as relations. The uniform layer now consists of ten machine-checked theorems (two corner, eight twin) covering 100% of the unconditional kills at all seven exponents tested. We conjecture, with this evidence, that these uniform theorems constitute the *complete* unconditional symbolic layer of the two-prime landscape at every exponent (a,b).

The machinery is not confined to two primes. Re-examining the thirty-six relations killed unconditionally at the three-prime exponent $(1,1,1)$: every one is a lone-plus-twin shape whose twin pair flips the conjugation of one or two primes' Gaussian factors. The twenty-eight single-flip relations are *literal instances* of the two-prime master theorems — the third prime's factor rides inside the opaque slot, its parity hypothesis discharged by closure of odd-real/even-imaginary parts under multiplication. The eight double-flip relations collapse over the composite $W = \bar{\chi}^4\psi^4$; the sum-type again lands in the master theorem with $w := W$, and the difference-type (lone slot $= \bar{W}$) is a short additional theorem resting on the fact that W is never real (star-fixedness would force $\bar{\chi} \mid \chi^4\bar{\psi}^4$, impossible for distinct split primes). The entire unconditional layer at $(1,1,1)$ is thus machine-checked by the same uniform toolkit.

Beyond the relation layer, the assignment-level Theorem E is now fully formal: **no magic square of squares has center entry** $(s\,p\,q)^2$ for distinct primes $p, q \equiv 1 \pmod 4$ and rigid cofactor s (`no_magic_square_of_squares_spq_center`, zero sorries). The proof pipeline: the machine-checked reduction produces four differences $u, v, u{+}v, u{-}v$ in the D-set; an eighteen-case Gaussian classification identifies every D-value as $\pm s^2 K$ with K one of four class magnitudes $\{q^2 \operatorname{Im}\pi^4,\ p^2 \operatorname{Im}\chi^4,\ \operatorname{Im}(\pi^4\chi^4),\ \operatorname{Im}(\pi^4\bar{\chi}^4)\}$; and a complete assignment analysis — six pair-dispatchers over three certificate tiers (two-term Gaussian regroupings with small odd coefficients, integer parity/product/size residues, and a coprimality-driven ratio system whose terminal quadratic factors as $(q^2-p^2)(q^2-2p^2)$) — kills all 1344 sign-and-class leaves. Together with Theorem C this makes the formally verified impossibility cover all center roots $s\,p^a$ and $s\,p\,q$.

The same program has now been carried one level deeper: **Theorem F is fully formal** — no magic square of squares has center entry $(s\,p^2q)^2$ (`no_magic_square_of_squares_sp2q_center`, zero sorries). Here the D-set splits into *seven* class magnitudes $\{p^4 \operatorname{Im}\chi^4,\ p^2q^2 \operatorname{Im}\pi^4,\ q^2 \operatorname{Im}\pi^8,\ p^2 \operatorname{Im}(\pi^4\chi^4),\ p^2$ I and the assignment analysis routes through sixteen buckets keyed by the number of level-eight slots: the L_2-free all-low bucket cancels a common p^2 and *reuses Theorem E verbatim*; lone level-eight and L_2 slots die by a p^2-extraction tier ($\pi \mid z$, $\pi \nmid \bar{z}$ forces $p \nmid \operatorname{Im} z$); and the six two-level-eight buckets reduce to two 25-cell cross-pair cores whose hardest cells include a valuation argument ending in $p^4 \mid 5R^4$ and coprime cancellation chains ending in factored quadratics such as $(p^4 - q^2)(p^4 - 2q^2)$. A new structural ingredient is a *ratio layer*: the pairwise distinctness of the four assigned class values reduces (through the branch $v = -2u$) to showing that no class value is twice another, a 49-cell analysis whose two deep cells — $\operatorname{Im}(\pi^4\chi^4) = \pm 2 \operatorname{Im}(\pi^4\bar{\chi}^4)$ and its level-eight analogue — factor a 3 through coprime chains and land on the norm systems $a^4 - b^4 = 8T^2$, i.e. the Fermat descent equation $u^4 - 4v^4 = w^2$. The formal proof resolves it by reduction to `not_fermat_42`: *the impossibility of a magic square of squares with center root $s\,p^2q$ thus rests, in part, on Fermat's right triangle theorem.* The verified frontier now covers all center roots $s\,p^a$, $s\,p\,q$, and $s\,p^2q$ — in particular every center that any Bremner-type near-miss square has ever exhibited. At the mixed exponent $(2,1,1)$ (1968 relations, 76 killed) two further composite phenomena appear and are absorbed: composite-lone shapes (lone slot $\pi^{4k}\chi^{4l}$ with twin part equal to the lone) rest on the fact that a product of positive quartic powers of distinct split primes is never real. At four primes the census of $(1,1,1,1)$ (14,944 relations, 232 killed) surfaces triple-composite twin shapes, absorbed by a final generalization: composite theorems with an *opaque tail* V (lone $= \overline{\chi^{4k}V}$, or lone $Z = \pi^{4k}V$ with twin Z^t), whose single hypothesis — the head prime does not divide $\bar{V}$ — is discharged mechanically for any number of further primes. The toolkit — corner, master-twin and arity-general composite theorems — covers every unconditional kill at all twelve exponent grids censused across two, three and four primes. Sharpness audits complete the characterization: every alive corner or twin shape fails a nameable hypothesis (fragile mixed pinch at anchor $\alpha \geq 2$, one-sided extraction closing only one prime ordering, sign-type mismatch with no common unit factor, composite lone slot, or the 2-adic violation), and zero alive relations satisfy the theorems' hypotheses.

Indeed the formal layer now *exceeds* the census engine. At $(2,2,1)$ the engine leaves 6208 relations alive, but 264 of them are corner triples with a constant spectator factor of the third prime: 80 have spectator power zero (the relation is p^4 times a two-prime corner, dead by the plain corner theorems), and 184 aligned-class relations carry $\psi^{4\gamma}$ spectators, killed by a spectator generalization (`uniform_corner_aligned_spec`) whose extraction chains stay clean because the aligned remainders group entirely on one side of the conjugation. The engine misses these because its per-prime certificate test cannot certify a two-prime pinch in the presence of a spectator. Notably, the attempted mixed-class spectator theorem *failed in Lean* for a real reason: the mixed χ-side core $e_3\pi^2 W - e_1\bar{\pi}^2\bar{W}$ mixes W with $\bar{W}$ and is no longer small, so those seventy-two relations are genuinely uncertified — the aligned/mixed asymmetry (robust versus fragile) is a theorem-level phenomenon, not an artifact.

The residual band relations were probed to second order: on several hundred integer points of the band loci the relation values carry $v_p = 1$ at 96% of points but $v_p = 2$ at about 4% — the mod-p^2 coefficient vanishes on a thinner sub-locus of its own, and each valuation order exhibits its own exceptional set. No bounded-valuation certificate is available by these methods: the last fraction of the two-prime $b = 2$ landscape is guarded by the same arbitrary-depth (Fermat-quotient–flavored) thinning as the deep order loci of §*Toward uniformity*, and its closure requires depth-uniform arithmetic input beyond first- and second-order valuations.

The grading lemma is exponent-independent, and its reach *grows* with the exponents: for general $e = s\,p^a q^b$ the set $D(e)$ is always $\{p^{2a-2\alpha}q^{2b-2\beta}s^2\ \mathrm{Im}(\pi^{4\alpha}\chi^{\pm 4\beta})\}$ of size $((2a+1)(2b+1)-1)/2$, and the census of leaves killed by the lemma reads 93.5% at $(2,2)$, 96.2% at $(3,2)$, 97.7% at $(3,3)$, and 98.9% at $(4,4)$, with the residual relation count growing only polynomially (216, 468, 1008, 3040). Thus the *entire* two-prime landscape reduces, uniformly in the exponents, to explicit doubly-balanced relation families carrying coupled-residue survival conditions.

Three primes

Nothing in the grading argument is special to two primes. For $e = s\,pqr$ (three distinct primes $\equiv 1 \pmod 4$) the set $D(e)$ consists of thirteen elements $p^{2-2\alpha}q^{2-2\beta}r^{2-2\gamma}s^2\ \mathrm{Im}(\pi^{4\alpha}\chi^{\pm 4\beta}\psi^{\pm 4\gamma})$ carrying a three-dimensional grading, and the lone-minimal-layer kill applies in any of the three gradings (68.4% of the 137,280 leaves). An exact sweep of the 43,392 residual leaves over *every* ordered triple of distinct such primes below 120 (94,768,128 checks) finds none satisfiable: *no magic square of squares has center root $s\,pqr$ with $p,q,r < 120$, for any s* — the first exclusion in the ≥ 3-prime landscape. The mixed shape $e = s\,p^2qr$ behaves the same way ($|D| = 22$, 86.9% of 1,404,480 leaves grading-killed, 92,413,440 exact checks, no survivor below 75). The grading lemma itself is machine-checked: `im_pipow_mul_not_dvd` in the accompanying Lean library proves, with no `sorry`, that for $p = a^2 + b^2$ prime, $k \geq 1$, and any Gaussian integer W with $p \nmid N(W)$, p never divides $\mathrm{Im}\big((a+bi)^k W\big)$. Four primes fall the same way: for $e = s\,pqrt$ ($|D| = 40$, four gradings), 81.2% of the 17,546,880 leaves die structurally and 396,656,640 exact checks exclude every quadruple from $\{5, 13, 17, 29, 37\}$ — no center root $s\,pqrt$ with all primes below 40. The "positive-dimensional wall" blocked only the solved-form pipeline; the raw-relation grading passes through it, and the same balanced-family characterization governs any number of primes.

For $e = s\,p^a q^b$ with two distinct primes $\equiv 1 \pmod 4$, the valuation pairs (ν_p, ν_q) organize $D(e)$ into *conjugate pairs* $\{d_{j,k}, d_{j,-k}\}$ of multiplicity exactly two off the axes, and axis singletons. If $\{u+v, u-v\}$ is a conjugate pair $\{d, d'\}$ arising from $A = \pi^{4j}$, $B = \chi^{4k}$, then

$$d + d' = 2\,\mathrm{scale} \cdot |\,\mathrm{Re}\,B\ \mathrm{Im}\,A|, \qquad d - d' = 2\,\mathrm{scale} \cdot |\,\mathrm{Im}\,B\ \mathrm{Re}\,A|,$$

so u and v are *completely determined* — and their valuation pair coincides with that of $\{d, d'\}$, forcing $u, v \in \{d, d'\}$, which is absurd. Consequently every hypothetical two-prime solution requires an explicit divisibility coincidence ($q^t \mid \mathrm{Re}\,\pi^{4j}$, $p^t \mid \mathrm{Re}\,\chi^{4k}$, or a mod-$p$ or mod-q cancellation in $u \pm v$). These coincidences recur along arithmetic progressions of exponents, so closing them requires exact-magnitude input beyond valuations; we leave this as the principal open direction, noting that together with Theorem 2 it would force every solution's center root into three or more primes $\equiv 1 \pmod 4$.

References

[1] C. Boyer, *Latest research on the "3x3 magic square of squares" problem*, `http://www.multimagie.com/English/SquaresOfSquaresSearch.htm`.

[2] A. Bremner, *On squares of squares II*, Acta Arith. 99 (2001), 289–308.

[3] M. Gardner, *The magic of 3x3*, Quantum 6 (1996), 24–26.